

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester III Examination, 2022-23

Paper No. BPT-301: Optics

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Symbols have their usual meanings.*

Question 1. Answer any **five** of the following:

[5 × 4 = 20]

- (a) An interference pattern from two point sources ($\lambda = 5000 \text{ \AA}$) separated by a distance $d = 0.05 \text{ mm}$ is produced on a screen PP' at a distance $D = 20 \text{ cm}$ from source S_1 . The plane of the screen PP' is perpendicular to the line joining S_1 and S_2 . What will be the shape of the fringes? What will be the order and the nature of the central fringe?
- (b) A left circularly polarized beam ($\lambda = 5893 \text{ \AA}$) is incident normally on a calcite crystal ($n_o = 1.65836$ and $n_e = 1.48641$) of thickness 0.005141 mm with its optic axis parallel to the surface. What will be the state of polarization of the emergent beam?
- (c) Discuss the state of polarization when the x - and y -components of the electric field are given by the following equations:

$$E_x = E_0 \sin(kz - \omega t + \pi/4), \quad E_y = E_0 \sin(kz - \omega t - \pi/4)$$

- (d) In a double-slit diffraction experiment, consider a slit width $b = 5 \times 10^{-2} \text{ cm}$ and a separation between identical points of the two slits $d = 0.1 \text{ cm}$, illuminated by a monochromatic light of wavelength $6 \times 10^{-5} \text{ cm}$. How many interference minima will occur within the central diffraction maximum?
- (e) A straight edge is kept midway between a monochromatic source ($\lambda = 5000 \text{ \AA}$) and a screen. The distance between the source and the screen is 50 cm . Calculate the distance of the first two maxima and the first two minima from the edge of the geometrical shadow.
- (f) Calculate the minimum number of lines in a grating that will resolve the sodium doublet ($\lambda_1 = 5890 \text{ \AA}$ and $\lambda_2 = 5896 \text{ \AA}$) in the second order.
- (g) Using sodium light of wavelength $\lambda = 5893 \text{ \AA}$, interference fringes are formed by reflection from a thin air wedge-shaped film. Ten fringes are observed in a distance of 1 cm when viewed perpendicularly. Calculate the angle of the wedge.

Question 2.

- (a) In the figure of Lloyd's mirror given below, S is a point source emitting waves of frequency $6 \times 10^{14} \text{ Hz}$, AB is the mirror and LOM is the screen. The distances SP , PA , AB and BO are 1 mm , 5 cm , 5 cm and 190 cm respectively. Determine the position of the region where the fringes will appear and the number of fringes in the region. [6]

- (b) Discuss the working of a Michelson interferometer and show that the highest order fringe appears at the centre. How can the difference between two close wavelengths of a bichromatic source be calculated using a Michelson interferometer? [8]
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Question 3.

- (a) Discuss the intensity distribution of Fraunhofer diffraction due to N slits. Deduce the expression for the width of the principal maxima. How many minima and secondary maxima are present between two closest principal maxima? [8]
- (b) In the Newton's ring arrangement, if the incident light consists of two wavelengths $4000 \text{ \AA}$ and $4002 \text{ \AA}$, calculate the distance from the point of contact at which the rings will disappear. The radius of curvature of the lens surface is 400 cm . [6]

OR

- (a) Derive the expression and discuss the intensity distribution of a Fabry-Pérot interferometer. Discuss how the sharpness of the fringes depends on the reflectivity of the mirrors. Give a brief comment on the resolving power of a Fabry-Pérot interferometer. [8]
- (b) Discuss the Rayleigh criterion of resolving power. Find the resolving power of a microscope. [6]
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Question 4.

- (a) Explain the working of a Retardation Plate, a Quarter-Wave Plate (QWP), and a Half-Wave Plate (HWP). [6]
- (b) Discuss the construction and working of a Nicol prism. [4]
- (c) What is optical activity? Explain Fresnel's theory of optical rotation. How is specific rotation experimentally determined using a Laurent half-shade polarimeter? [4]

OR

- (a) The plane of polarization of a plane-polarized light is turned through an angle 12.6° passing through a 10% sugar solution of length 20 cm . Calculate the specific rotation. [6]
- (b) What are Einstein coefficients? Derive the relation between them. [8]
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Question 5.

- (a) Discuss the intensity distribution due to a single-slit diffraction and find the width of the central maximum. [14]
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